# Company Introduction

MediLab Diagnostics is a growing laboratory network providing testing services to individuals, hospitals, clinics, HMOs and corporate healthcare partners across Nigeria.

It operates multiple branches and performs tests in haematology, clinical chemistry, microbiology, serology, hormonal assays, molecular diagnostics and histopathology.

Information on patients, tests, branches, referring clinicians, outcomes and finances is stored in a single flat file. Management wants a clearer view of diagnostic volume, patient activity, branch performance, turnaround time and financial outcomes.

**8**
Branches

**7**
Test Categories

**20**
Different Tests

**10**
Referring Clinicians

Why this analysis was needed

# The Business Problem

## One flat file

Patient, branch, clinician and test details repeat across many records, so the same information is stored again and again.

## Hard to trust and maintain

Duplicates, inconsistent spellings and mixed formats make calculations unreliable and reports hard to update when new data arrives.

## The insight bottleneck

Leadership needs visibility of their data and a faster turnaround for reporting on volume, branch performance, services and finances.

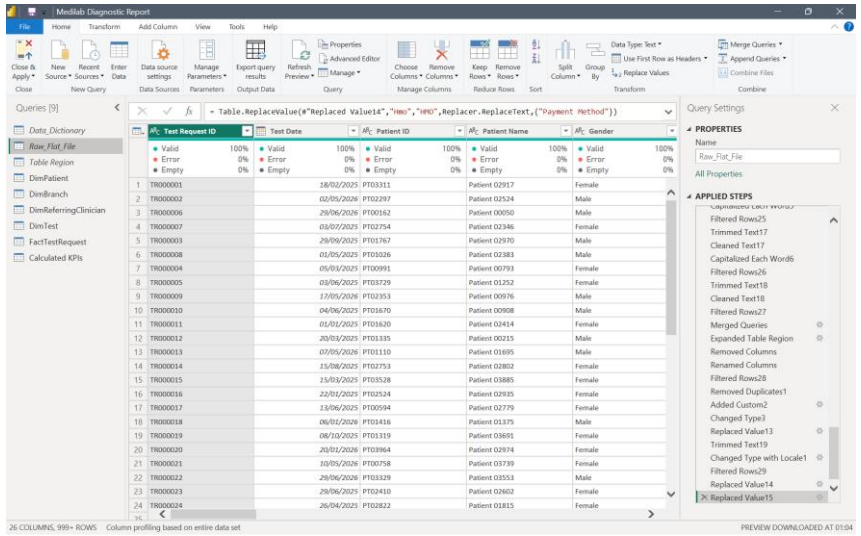
Goal: turn the flat file into a clean, structured data model and a two-page dashboard that answers management's questions on demand, service mix, branch performance, operations and profitability.

# Data Cleaning: Faults Found and Fixed

| Fault found | Fix applied |
|---|---|
| Duplicate requests | 25 repeated Test Request IDs removed (Remove Duplicates) |
| Gender: 6 spellings | Trim, Clean, Capitalize Each Word: Female / Male |
| Age Group: 10 spellings (e.g., "0 to 17") | Replace Values: 0-17, 18-30, 31-45, 46-60, 60+ |
| Branch Name 20 variants, Customer Type 11, Payment Method 11, Test Status 8, Specialty 10 | Trim, Clean, Replace Values: 8 branches, 5 customer types, 5 payment methods, 3 statuses, 5 specialties |
| ID columns with leading spaces | Trim and Clean (Patient IDs 3,808 become 3,789) |
| Region: "Lagos" used as a region, South West spelt 3 ways | Region lookup table merged on State: one region per state |
| Numbers stored as text (commas, " days") | Replace Values, then Change Type with Locale to decimal |
| 30 dates stored as text in mixed formats | Converted to Date type: no blanks or errors remain |

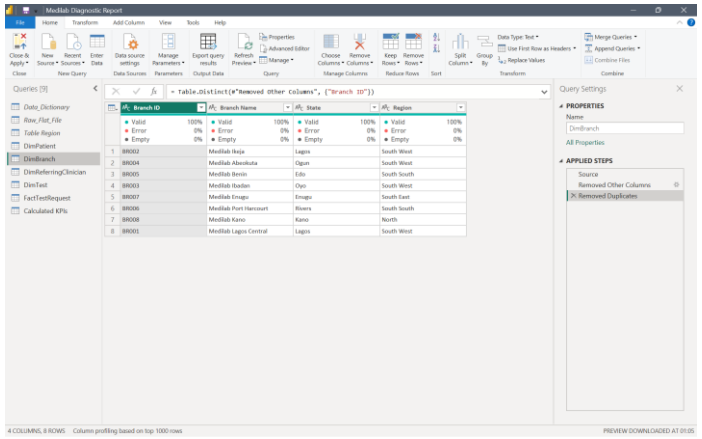
**12,025**
rows before (26 columns)

**12,000**
rows after - one row per test request

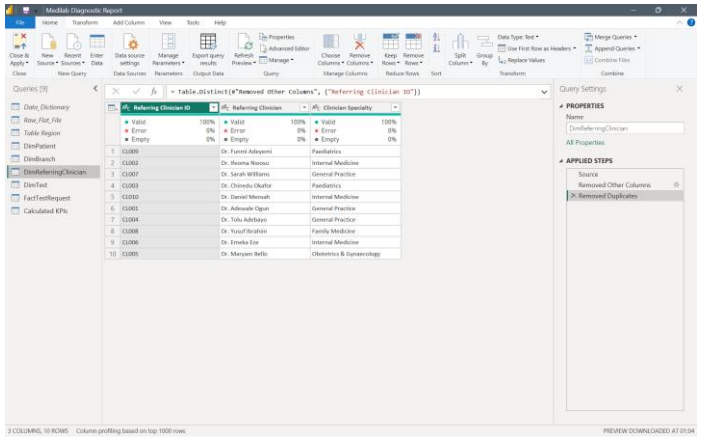


Applied steps in Power Query (Raw_Flat_File)

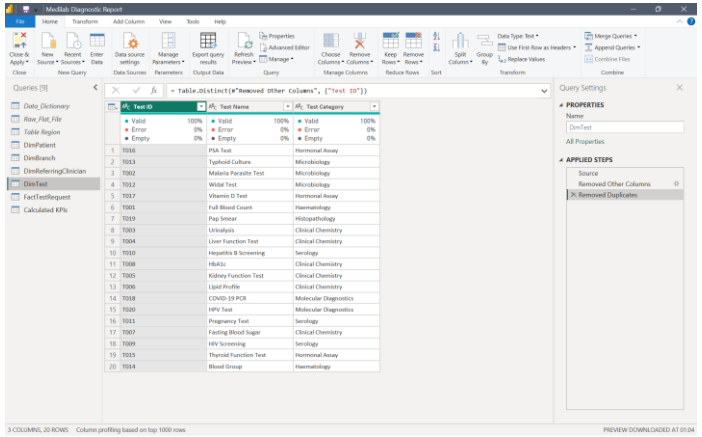
# Cleaned Dimension Tables in Power Query



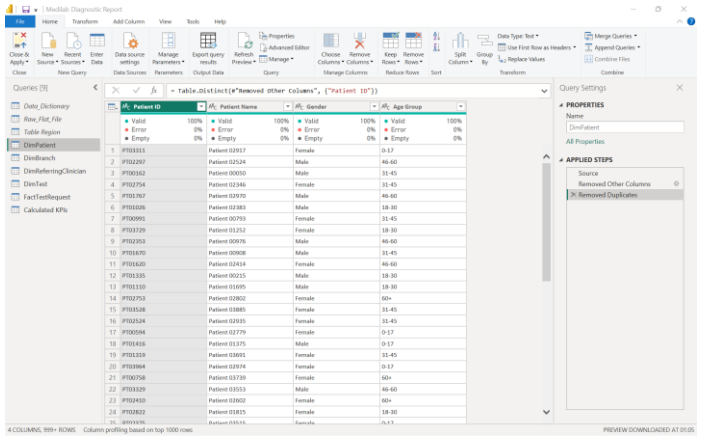
**DimBranch: 8 rows**



**DimReferringClinician: 10 rows**



**DimTest: 20 rows**



**DimPatient: 3,789 patients**

Each dimension keeps only its own columns and is de-duplicated on its ID (Remove Duplicates), so every key is unique. The fact table holds 12,000 rows.

# Data Quality Trends Found and Decisions Taken

**17**

## cancelled requests still carry revenue

₦150.2K revenue and ₦62.4K cost sit on cancelled requests, mostly Kano (₦44.6K) and Abeokuta (₦30.5K). Kept as recorded and flagged with a Cancelled With Revenue column so the amount can be reported or excluded.

**4**

## completed requests show ₦0 revenue and cost

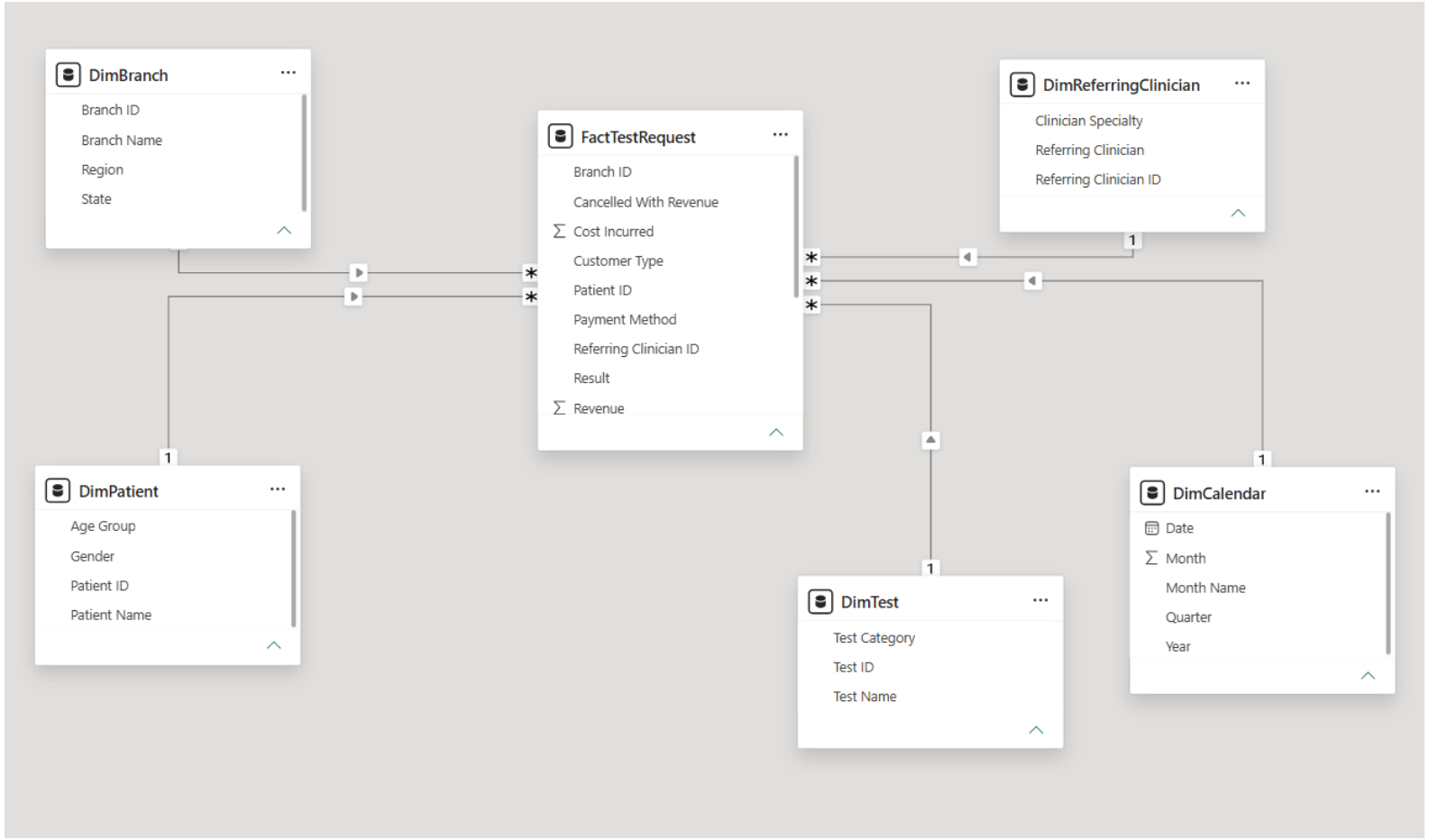
Worth ₦33.5K at list price (0.04% of revenue). Completed tests should earn their price, so this looks like an entry fault. Left unchanged to preserve the original entries, and reported as a finding (TR004017, TR004685, TR010607, TR010852).

**2**

## pending requests show ₦0 revenue

The other 408 pending requests show revenue (374 at 15% of price, 34 at full price): ₦754.8K in total. Pending revenue was kept as recorded and noted as a rule to confirm.

# Data Model: Star Schema




## How to read it

The centre table records every test request. Around it, dimension tables describe who (DimPatient, DimReferringClinician), where (DimBranch), what (DimTest) and when (DimCalendar).

Each link is one-to-many (1 to *), single direction: one branch, patient, clinician, test or date matches many requests.

One row per request, so revenue and cost are counted once and never repeated.

# Key Measures and Calculations

| Measure | Definition, and what it means in plain words |
|---|---|
| **Total Test** | Distinct count of Test Request ID, meaning how many separate test requests were made. |
| **Total Patient** | Distinct count of Patient ID, meaning how many different people used the lab (each person counted once). |
| **Revenue / COGS** | Sum of Revenue / sum of Cost Incurred, meaning the money earned from tests / the direct cost of carrying them out (cost of goods sold). |
| **Profit** | Revenue minus COGS, meaning the money left after paying the direct costs of the tests. |
| **Profit Margin** | Profit divided by Revenue, meaning how much profit is kept from every naira earned. |
| **Avg. Turnaround (Days)** | Average of Turnaround Time (Days), meaning the typical number of days a patient waits from request to result. |
| **Completed Test Rate** | Completed requests divided by all requests, meaning the share of requests that end in a finished result. |
| **Revenue per Patient** | Revenue divided by the number of distinct patients, meaning the average value of one patient. |
| **Tests per Patient** | Total Test divided by Total Patient, meaning how many tests the average patient takes. |

# Network Activity at a Glance

**Business question:** *How active has MediLab's diagnostic network been during the period under review? The Page 2 KPIs (Revenue per Patient, Tests per Patient, Completed Test Rate) are shown with it.*

Page 1 KPIs

| 12,000 | 3,789 | ₦91.16M | ₦38.08M | ₦53.08M | 58.22% | 1.04 |
|---|---|---|---|---|---|---|
| Total Test | Total Patient | Revenue | COGS | Profit | Profit Margin | Avg. Turnaround (Days) |

Page 2 KPIs

| ₦91.16M | ₦53.08M | 58.22% |
|---|---|---|
| Revenue | Profit | Profit Margin |
| ₦24.06K | 3.17 | 90.13% |
| Revenue per Patient | Test Per Patient | Completed Test Rate |

## Insight

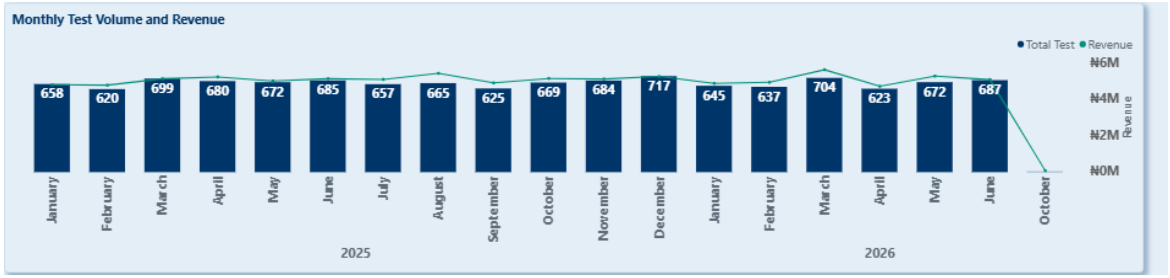
Between 1 January 2025 and 2 October 2026, MediLab handled 12,000 test requests from 3,789 different patients, bringing in ₦91.16M. After ₦38.08M in direct costs (COGS), ₦53.08M remained as profit, about 58 kobo of every naira earned. Results take about one day (1.04). On average each patient was worth ₦24.06K and took 3.17 tests, and 90.13% of requests were completed.

## Recommendation

Use these KPIs as the monthly scorecard: a 58% margin or better, results within 1 day and a completed rate above 90%.

# Monthly Test Volume and Revenue

**Business question:** *How has diagnostic activity and its financial contribution evolved over time?*




## Insight

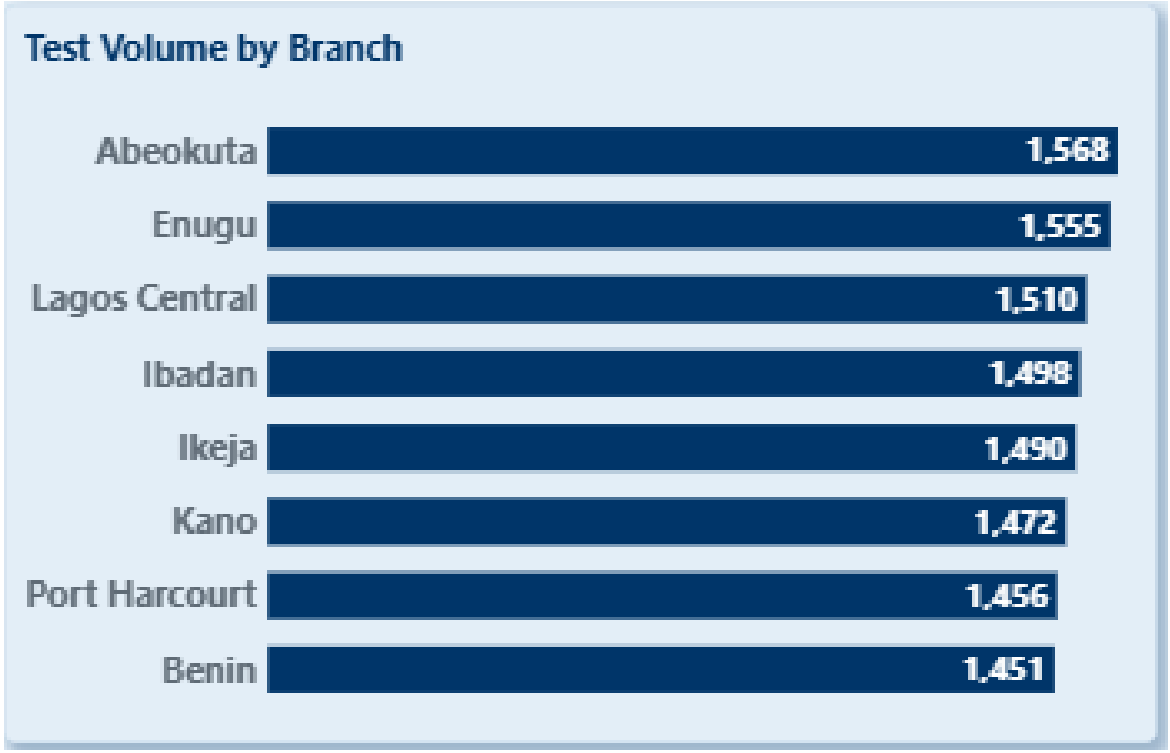
Demand is steady month to month: each month has 620 to 717 tests and ₦4.7M to ₦5.6M in revenue, so there is no busy or quiet season. March 2026 earned the most (₦5.6M) even though December 2025 had more tests (717 against 704), because higher-priced tests were done. October 2026 drops to zero only because the data stops on 2 October.

## Recommendation

Volume is flat at about 650 tests a month, so grow revenue through new referrals and higher-value tests such as Molecular Diagnostics.

# Test Volume by Branch

**Business question:** *Where is the greatest concentration of diagnostic activity across MediLab's network?*




## Insight

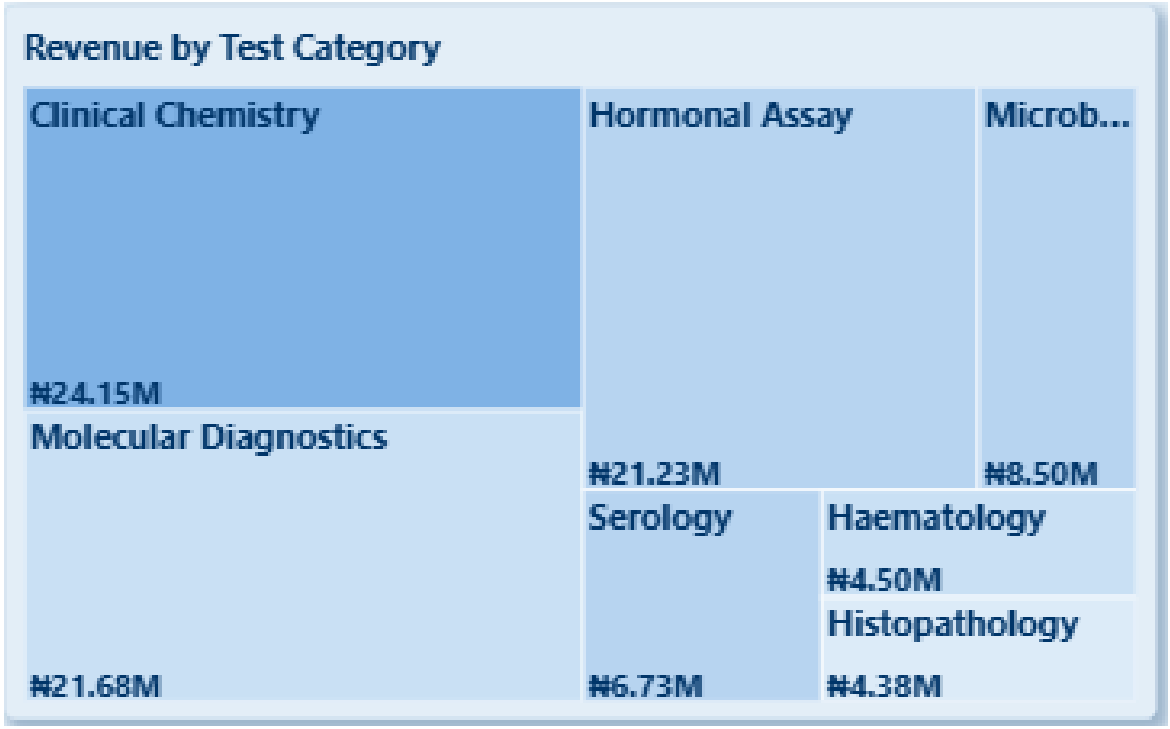
Work is shared almost equally across the eight branches, each handling between 1,451 and 1,568 tests. Abeokuta is the busiest with 1,568 and Benin the quietest with 1,451, a difference of only 117 tests. In plain terms, no single branch carries the network, so volume alone does not separate strong branches from weak ones.

## Recommendation

No branch is over-relied on (1,451 to 1,568 tests each), so judge branches on margin, completion and turnaround. Give Abeokuta (1,568) and Enugu (1,555) capacity first.

# Revenue by Test Category

**Business question:** *Which diagnostic services are contributing most significantly to the company's revenue?*




## Insight

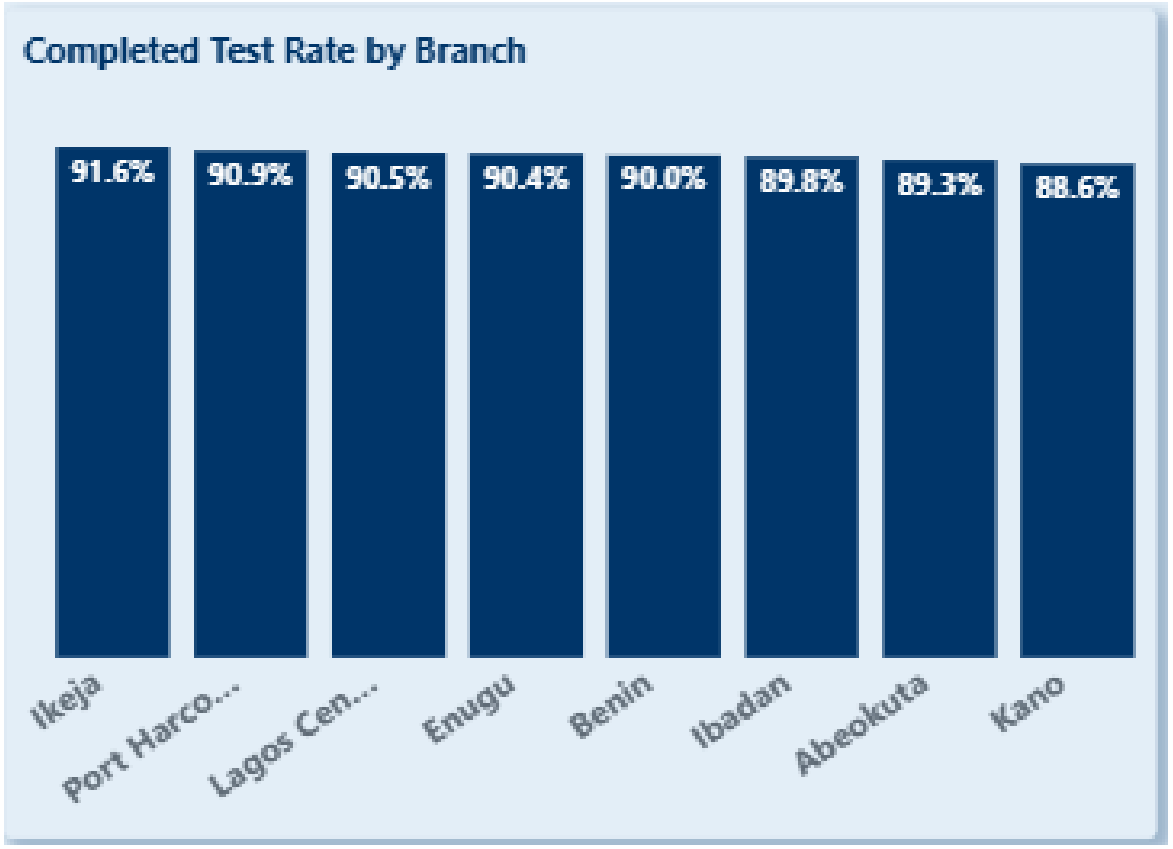
Revenue depends heavily on three services. Clinical Chemistry (₦24.15M), Molecular Diagnostics (₦21.68M) and Hormonal Assay (₦21.23M) together bring in 73.6% of all revenue. Histopathology is the smallest at ₦4.38M. If these three services were disrupted, most of MediLab's income would be affected.

## Recommendation

Protect supplies and equipment for the three categories that earn 73.6% of revenue, and grow the smaller high-margin lines, Serology and Haematology.

# Completed Test Rate by Branch

**Business question:** *How efficiently is MediLab turning test requests into completed diagnostic services?*




## Insight

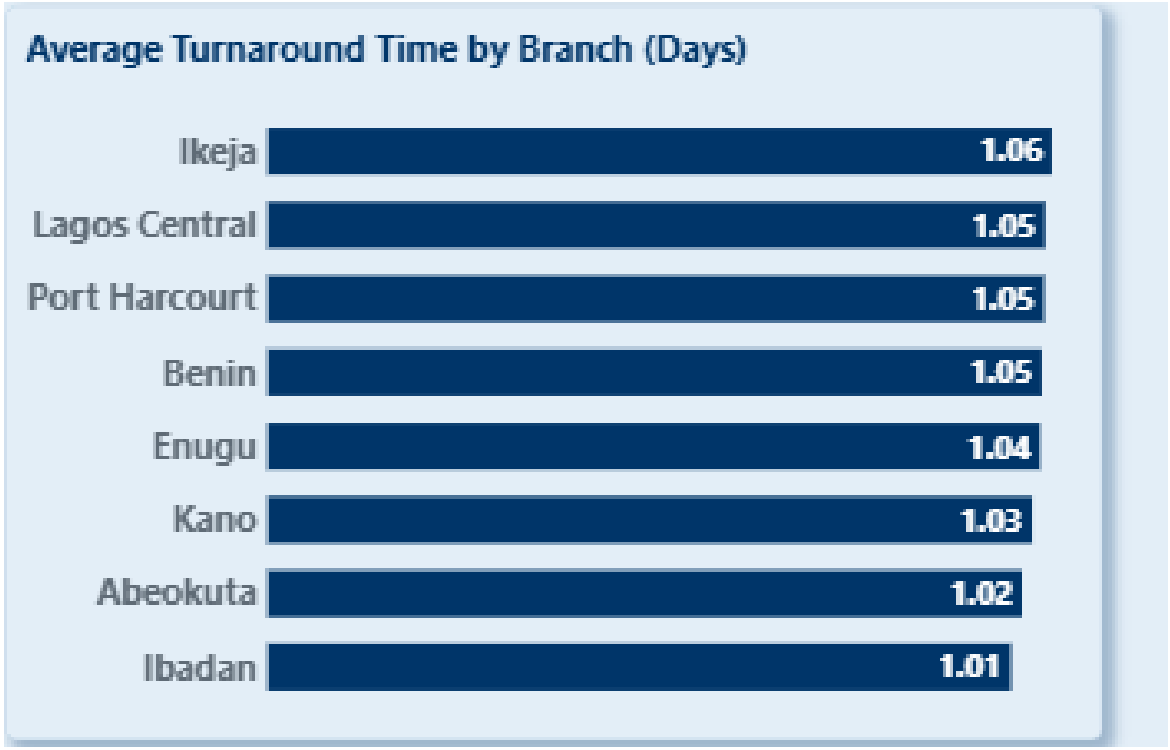
Completion rate is the share of requests that end in a finished result. Across the network it is 90.13%, so roughly 9 in every 10 requests are completed. Ikeja does best (91.6%) and Kano least well (88.6%), with Abeokuta also below average at 89.3%. The chart starts at 80%, so these small gaps are easy to see.

## Recommendation

Find out why Kano and Abeokuta lose more requests. Lifting Kano from 88.6% to the 90.1% average would complete about 23 more tests.

# Average Turnaround Time by Branch (Days)

**Business question:** *Where are operational delays most evident across the branch network?*




## Insight

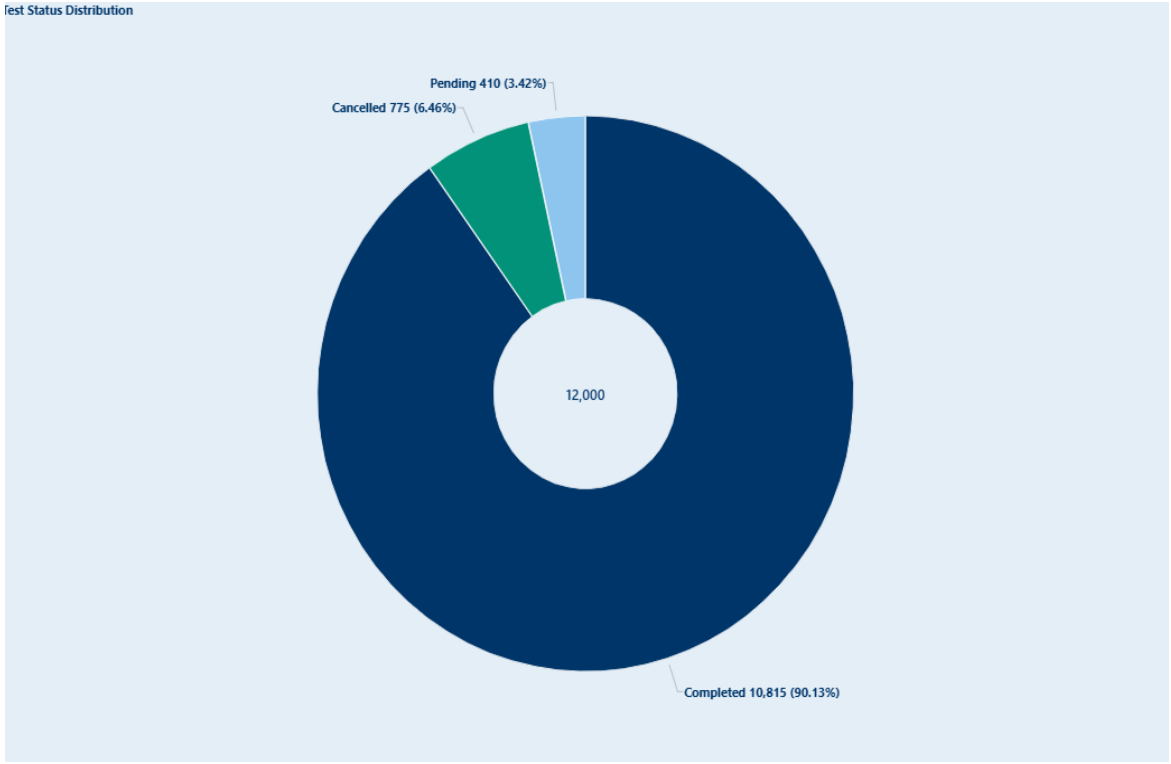
Turnaround time is how long a patient waits for results. Every branch is about the same (1.01 to 1.06 days), so the branch is not the problem. The type of test is, Histopathology takes about 3 days and Molecular Diagnostics 2 days, while Haematology and Clinical Chemistry take only 0.4 to 0.5 days.

## Recommendation

Aim improvement at Histopathology (3.0 days) and Molecular Diagnostics (2.0 days), for example batching samples or adding staff, rather than at one branch.

# Test Status Distribution

**Business question:** *What does the distribution of completed, cancelled and pending requests tell us about operational performance?*




## Insight

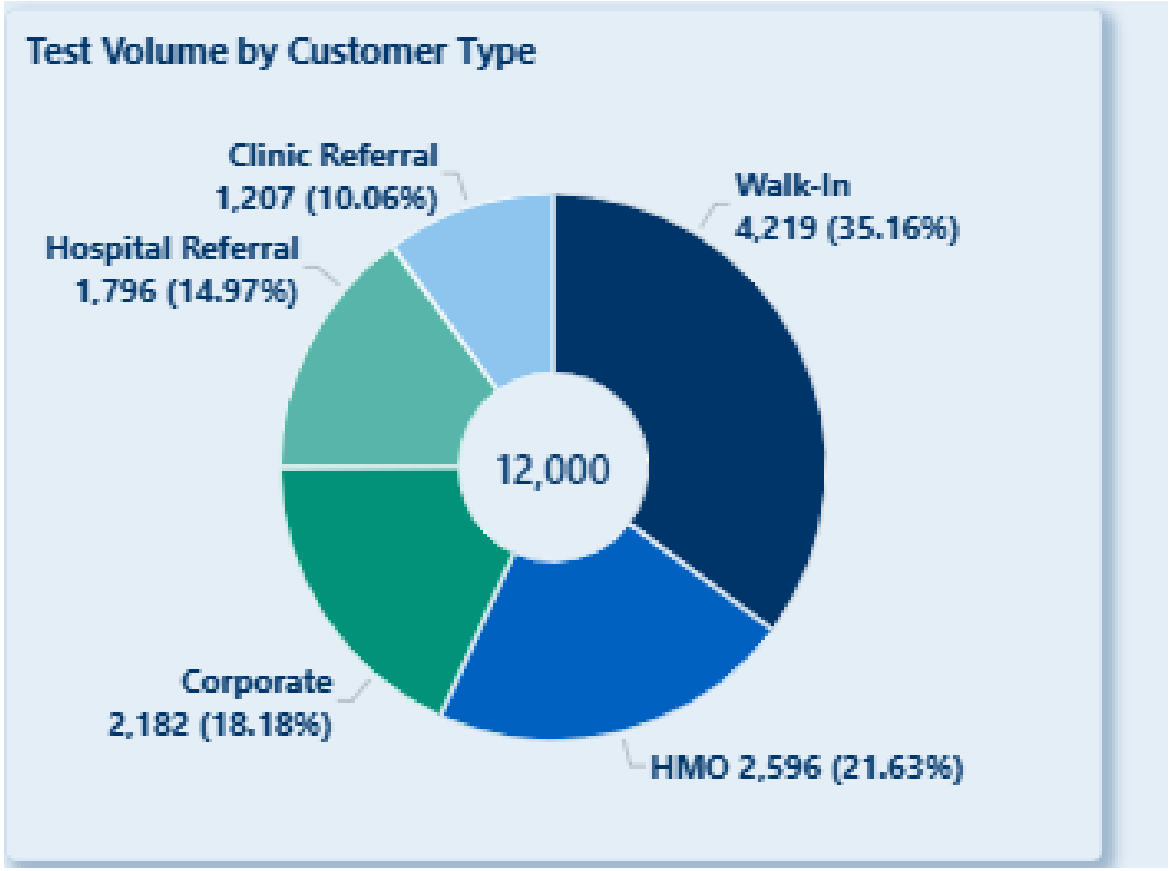
Out of 12,000 requests, 10,815 (90.13%) were completed, 775 (6.46%) were cancelled and 410 (3.42%) are still pending. So about 1 in 10 requests does not end in a result. Cancellations are almost twice as common as pending cases, which may point to requests being dropped rather than waiting in a queue.

## Recommendation

Record a cancellation reason and follow up pending requests within 24 hours to recover part of the 1,185 requests (9.9%) not completed.

# Test Volume by Customer Type

**Business question:** *Which customer groups are driving the greatest demand for MediLab's services?*




## Insight

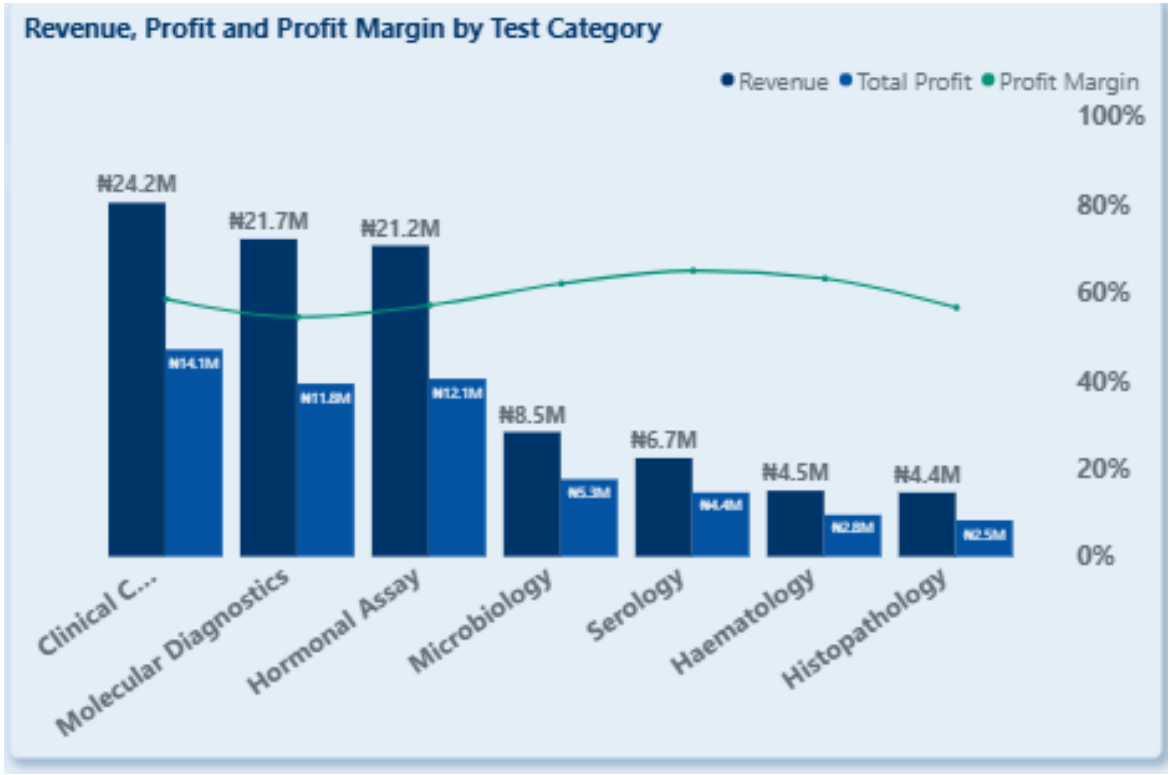
Walk-in patients are the biggest source of demand with 4,219 tests (35.16%), followed by HMOs (2,596; 21.63%) and corporate clients (2,182; 18.18%). Referrals from hospitals and clinics add 3,003 tests (25%). Over a third of the business therefore depends on people who choose MediLab themselves rather than on contracts or referrals.

## Recommendation

Protect the walk-in experience (35% of volume) at each branch and sign more HMO and corporate contracts for predictable volume.

# Revenue, Profit and Profit Margin by Test Category

**Business question:** *Which diagnostic services are making the greatest contribution to MediLab's business? Which areas of the service portfolio deliver both strong profitability and healthy margins?*




## Insight

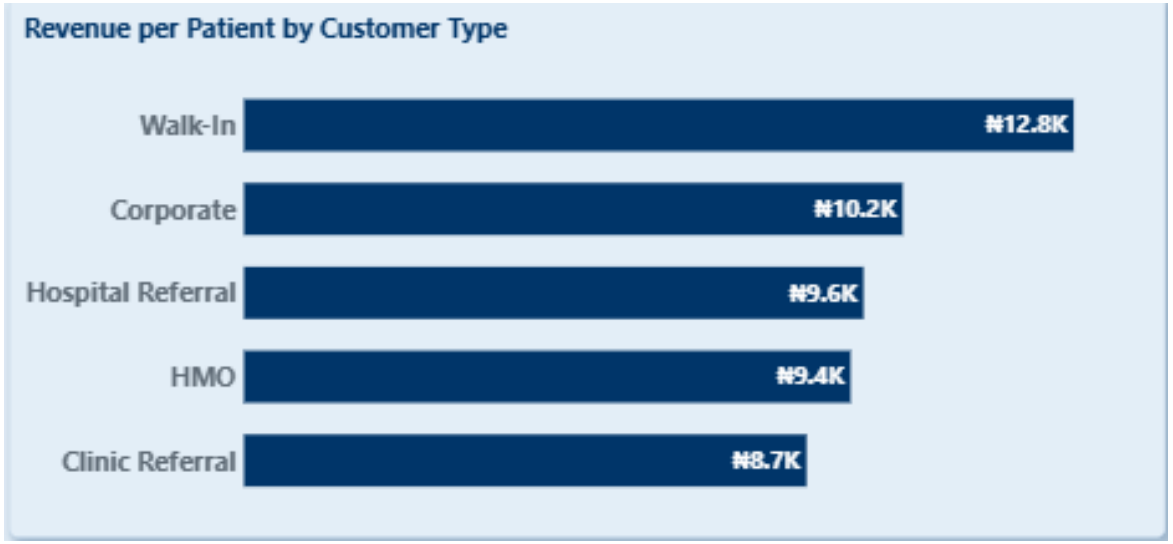
Clinical Chemistry is the biggest earner (₦24.15M revenue, ₦14.1M profit). Profit margin is the share of revenue kept as profit: Serology keeps the most (65.1%) and Haematology 63.3%, but both earn little overall. Molecular Diagnostics is the second-largest earner yet keeps the least (about 54.5%), because more of its revenue goes to costs.

## Recommendation

Grow Serology and Haematology volume and review Molecular Diagnostics pricing and costs: a 58% margin there would add about ₦0.8M profit.

# Revenue per Patient by Customer Type

**Business question:** *Which customer groups are generating the greatest financial value for MediLab relative to the patients they serve?*




## Insight

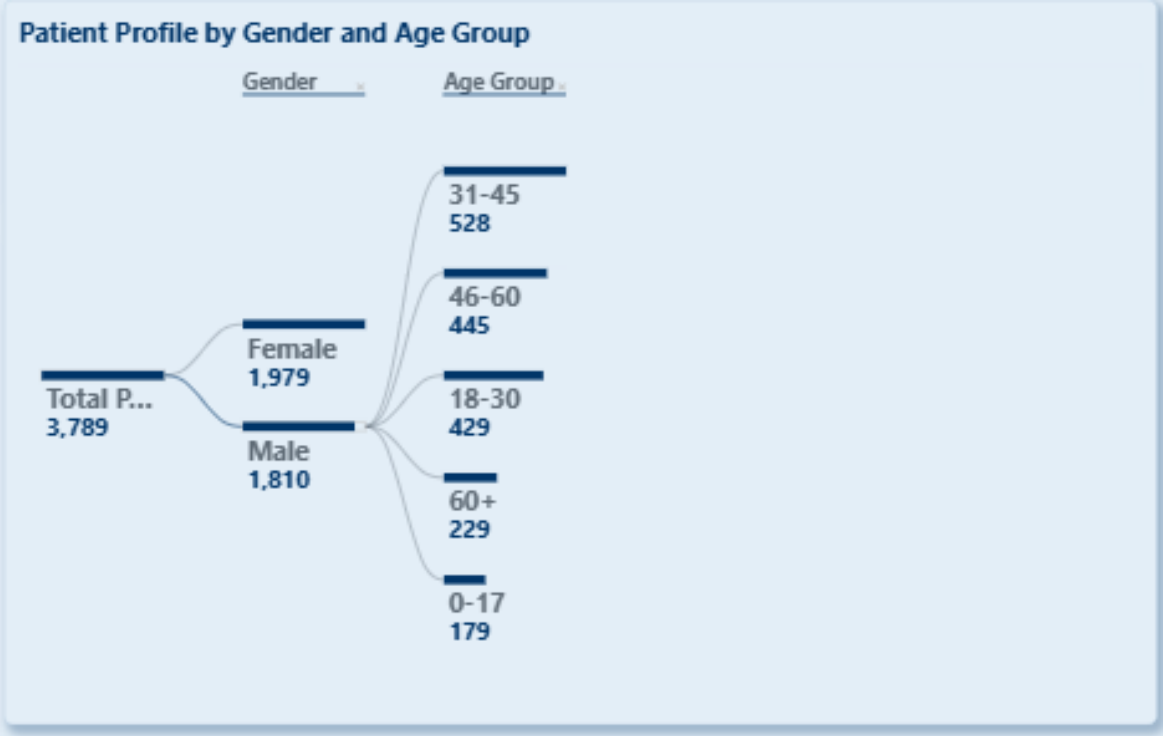
Revenue per patient shows what each patient is worth on average. Walk-ins are worth the most (about ₦12.8K), then Corporate (₦10.2K), Hospital Referral (₦9.6K), HMO (₦9.4K) and Clinic Referral (₦8.7K). Walk-ins also come back more often, with 1.6 tests per patient compared with 1.1 for clinic referrals, which is why they are worth more.

## Recommendation

Invest in walk-in retention such as repeat-visit reminders (₦12.8K per patient), and grow corporate accounts, the second most valuable group at ₦10.2K.

# Patient Profile by Gender and Age Group

**Business question:** *What does the composition of MediLab's patient population look like, and are there notable differences across demographic groups?*




## Insight

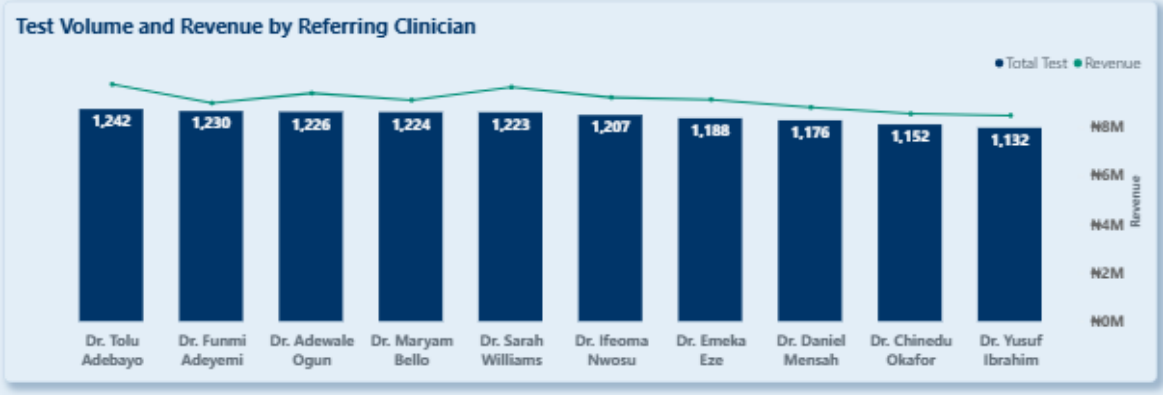
MediLab has 3,789 patients, slightly more female (1,979, or 52.2%) than male (1,810, or 47.8%). The age pattern is almost the same for both: 31 to 45 is the largest group (552 women, 28%; 528 men, 29%) and 0 to 17 the smallest (195 women and 179 men, about 10% each). About 77% of both women and men are working-age adults (18 to 60). Women outnumber men in every age group, most clearly at 46 to 60 (513 against 445).

## Recommendation

Design screening packages for the 31 to 45 and 46 to 60 age groups (2,038 patients across both genders) and consider women's health bundles for the larger female base (1,979).

# Test Volume and Revenue by Referring Clinician

**Business question:** *Which referring clinicians are driving the greatest volume of diagnostic activity, and which are contributing most to revenue?*




## Insight

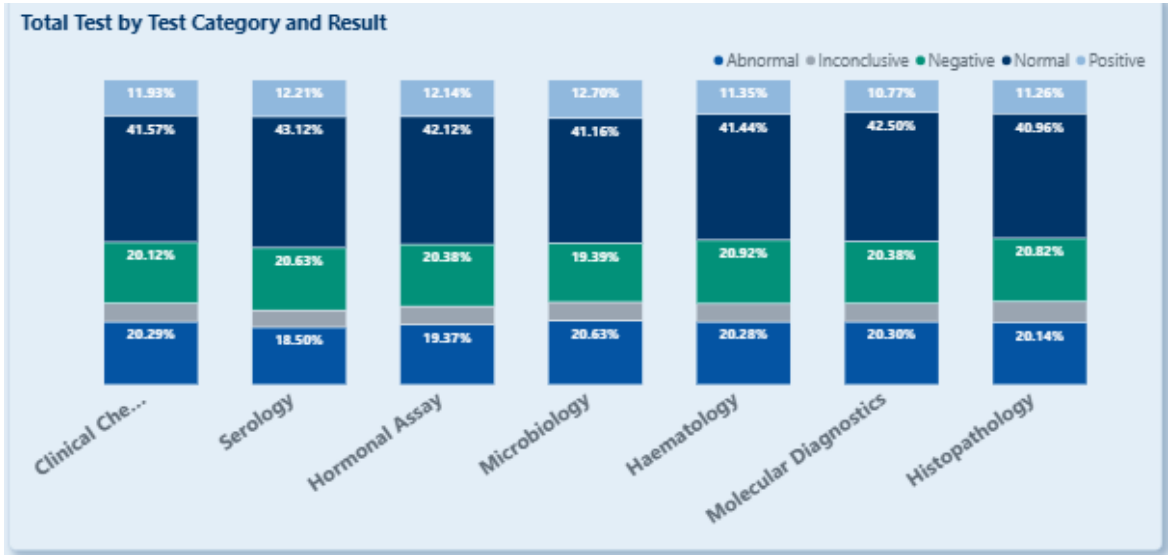
Referring clinicians are the doctors who send patients to MediLab. Dr. Tolu Adebayo sends the most tests (1,242) and brings in the most revenue (about ₦9.8M); Dr. Yusuf Ibrahim sends the fewest (1,132 tests, about ₦8.5M). Test counts differ by only 9.7% across the ten doctors, but revenue differs by 15%, so the type of tests ordered matters as much as how many.

## Recommendation

Share the high-value test menu (Molecular, Hormonal) with lower-revenue referrers such as Dr. Ibrahim (₦8.5M against ₦9.8M) to lift revenue without more volume.

# Test Results by Test Category

**Business question:** *What patterns can be observed in diagnostic results across the different test categories?*




## Insight

Results are strikingly similar in every category: 41% to 43% are Normal and 18.5% to 20.6% Abnormal. Serology has the highest share of normal results (43.1%) and the lowest of abnormal (18.5%). No category shows an unusual pattern, so no test type is producing an unexpected number of abnormal or inconclusive results.

## Recommendation

Apply one network-wide quality standard and track the 722 inconclusive results (6% of tests) for repeat testing.

# Branch Performance Scorecard

**Business question:** *Which branches are successfully balancing diagnostic volume with financial performance? Which areas appear to require management attention, and what evidence supports your conclusion?*

**Volume, Profit Margin, Completion and Turnaround by Branch**

| Branch Name | Total Test | Revenue | Profit Margin | Completed Test Rate | Average Turnaround Time | Cancelled Revenue |
|---|---|---|---|---|---|---|
| Abeokuta | 1,568 | ₦11.96M | 58.02% | 89.29% | 1.02 | ₦30.54K |
| Benin | 1,451 | ₦10.81M | 58.20% | 90.01% | 1.05 | ₦0.00K |
| Enugu | 1,555 | ₦11.93M | 58.17% | 90.35% | 1.04 | ₦4.75K |
| Ibadan | 1,498 | ₦11.25M | 58.31% | 89.79% | 1.01 | ₦22.00K |
| Ikeja | 1,490 | ₦11.28M | 58.40% | 91.61% | 1.06 | ₦24.23K |
| Kano | 1,472 | ₦10.83M | 58.49% | 88.59% | 1.03 | ₦44.63K |
| Lagos Central | 1,510 | ₦11.88M | 57.99% | 90.46% | 1.05 | ₦4.91K |
| Port Harcourt | 1,456 | ₦11.22M | 58.24% | 90.93% | 1.05 | ₦19.18K |

## Insight

The branches are well balanced: each handles 1,451 to 1,568 tests with margins between 57.99% and 58.49%. Kano needs attention, with the lowest completion rate (88.59%) and the most revenue recorded on cancelled requests (₦44.63K), followed by Abeokuta (89.29%, ₦30.54K). Revenue on cancelled requests may mean refunds were not recorded.

## Recommendation

Review cancellations at Kano and Abeokuta first and check for unrecorded refunds: ₦150.2K sits on 17 cancelled requests network-wide.

# Summary of Recommendations

MediLab DIAGNOSTICS

1

## Close the completion gap

1,185 requests (9.9%) are cancelled or pending. Start with Kano (88.6%) and Abeokuta (89.3%) and add cancellation reasons.

2

## Audit cancelled revenue

17 cancelled requests carry ₦150.2K. Add a refund status so reports show true revenue.

3

## Fix the margin mix

Molecular Diagnostics earns 23.8% of revenue at about 54.5% margin; Serology reaches 65.1%.

4

## Speed up slow tests

Histopathology (3.0 days) and Molecular Diagnostics (2.0 days) are far slower than other tests (0.4 to 1.5 days).

5

## Grow high-value channels

Walk-ins are 35% of volume and worth about ₦12.8K per patient; add HMO and corporate contracts.

6

## Tighten data capture

4 completed requests and 2 pending requests show ₦0, and patient details differ across the same Patient ID.

# Final Dashboard: Both Pages

